

Sprint 5 Deliverable

University Inventory Management System

Srinidh & Jithender

Sprint Number:	Sprint 5
Sprint Duration:	2 Weeks (Week 9 – Week 10)
Sprint Goal:	Implement the Reporting and Analytics module, complete system and acceptance testing across all modules, fix all critical bugs, and deploy the application.
Team:	Srinidh, Jithender
Course:	Software Engineering Lab
Date:	April 2026

1. Sprint Plan & Backlog

1.1 Sprint Goal

Deliver the final sprint: Admins can generate reports (room-wise inventory summary, verification status, overdue items), export them as CSV, and view a dashboard with key metrics and charts. The team completes end-to-end system testing, fixes all outstanding bugs, writes final documentation, and prepares the application for deployment. This sprint covers UC-009 (Generate Report) and UC-010 (View Dashboard).

1.2 Sprint Backlog

The following user stories and tasks are selected from the product backlog for Sprint 5.

ID	User Story / Task	Priority	Est. Hrs	Status
US-22	As an Admin, I want a dashboard showing total items, verified items, and overdue items.	High	4	To Do
US-23	As an Admin, I want to generate a room-wise inventory report filtered by department.	High	4	To Do
US-24	As an Admin, I want a verification status report showing Verified and Pending items.	High	3	To Do
US-25	As an Admin, I want to export any report as a CSV file.	Medium	3	To Do
US-26	As an Admin, I want to see items that have not been verified in the last 6 months.	Medium	3	To Do
T-23	Build dashboard page with summary cards and Chart.js bar/pie charts.	High	5	To Do
T-24	Implement /api/reports endpoints (inventory summary, verification status, overdue).	High	5	To Do
T-25	Implement CSV export for each report type.	Medium	2	To Do
T-26	Execute full system test scripts for all 10 use cases end-to-end.	High	4	To Do
T-27	Conduct acceptance testing with simulated Admin and VO user sessions.	High	3	To Do
T-28	Fix all P1 and P2 bugs found during system testing.	High	4	To Do

T-29	Write final README, API documentation, and deployment guide.	Medium	3	To Do
T-30	Deploy application to a production-ready environment.	Medium	3	To Do

1.3 Sprint Timeline

Day	Activity
Day 1–2	Implement /api/reports endpoints; compute 180-day overdue threshold
Day 3–4	Build dashboard UI with summary cards and Chart.js charts (live API data)
Day 5	Add CSV export for inventory and verification reports
Day 6–7	Implement overdue items report; handle NULL ver_date edge case
Day 8–10	Full system testing: execute test scripts for UC-001 through UC-010
Day 11	Acceptance testing with simulated Admin and VO sessions
Day 12	Bug fix sprint: address all P1 and P2 issues from testing
Day 13	Write README, API documentation, and deployment guide
Day 14	Final deployment, sprint review, project retrospective

1.4 Definition of Done

- Dashboard shows accurate counts: total items, verified this cycle, pending, and overdue.
- Room-wise inventory report lists all items per room with current status.
- Verification status report shows each item's last verified date and status.
- Overdue report correctly lists items not verified in the last 180 days.
- CSV export downloads correctly for all report types.
- All 10 use cases pass system test scripts with no P1 defects remaining.
- Acceptance testing sign-off obtained for both Admin and VO simulated sessions.
- README documents project setup, environment variables, and run instructions.
- Application runs successfully in production build (node server.js).

1.5 Sprint Risks

Risk	Likelihood	Impact	Mitigation
Overdue query returns incorrect results for items never verified (NULL ver_date).	Medium	High	Handle NULL explicitly: treat items older than 180 days with no verification as overdue.
CSV encoding issues with special characters in item names.	Low	Low	Write UTF-8 BOM to CSV output for Excel compatibility.
Chart.js renders slowly on large datasets (500+ items).	Low	Medium	Aggregate data server-side; send chart-ready JSON rather than raw rows.
Deployment environment differences (Node version, MySQL config).	Medium	High	Document exact versions in README; test on a clean environment before submission.

2. Reporting Module Design

2.1 Use Cases Addressed

Use Case	Description
UC-009	Admin generates an inventory or verification status report.
UC-010	Admin views the real-time dashboard with summary metrics and charts.

2.2 API Endpoints

Method	Endpoint	Description
GET	/api/reports/inventory	Room-wise item count; optional ?dept_id= filter.
GET	/api/reports/verification	Item verification status (Verified / Pending).
GET	/api/reports/overdue	Items not verified in the last 180 days.
GET	/api/reports/inventory/csv	CSV download of the inventory report.
GET	/api/reports/verification/csv	CSV download of the verification report.
GET	/api/dashboard	Aggregated metrics for dashboard cards and charts.

2.3 Overdue Items SQL Query

```
SELECT i.item_id, i.item_name, r.room_name,  
MAX(v.ver_date) AS last_verified  
FROM ITEM i  
JOIN ROOM r ON r.room_id = i.room_id  
LEFT JOIN VERIFICATION v  
ON v.item_id = i.item_id AND v.status = 'Verified'  
WHERE i.status = 'Active'  
GROUP BY i.item_id, i.item_name, r.room_name  
HAVING last_verified IS NULL  
OR last_verified < DATE_SUB(NOW(), INTERVAL 180 DAY);
```

2.4 Dashboard Layout

The dashboard page (HTML/CSS/JS with Chart.js) shows four summary cards: Total Items, Verified This Cycle, Pending Verification, and Overdue Items. Below the cards, two Chart.js charts render: (1) a bar chart of item count per department, and (2) a pie chart of item status distribution (Active vs Decommissioned). All data is fetched from GET /api/dashboard on page load.

3. System Test Plan

3.1 Test Scope

All 10 use cases (UC-001 through UC-010) are covered by system test scripts. Each script simulates a realistic user session using Supertest (Node.js) and a manual browser walkthrough.

3.2 System Test Cases

Test ID	Use Case	Steps (summary)	Expected Result
ST-01	UC-001	Login with valid Admin credentials.	Dashboard loads; role shown as Admin.

ST-02	UC-001	Login with incorrect password.	Error message: Invalid credentials.
ST-03	UC-002	Add department 'Physics'; attempt to add duplicate.	First succeeds (201); duplicate returns 409.
ST-04	UC-003	Add room 'Lab 101' in Physics; view room list.	Room appears in the filtered department list.
ST-05	UC-004	Add item 'Desktop PC' in Lab 101.	Item created; QR code displayed on screen.
ST-06	UC-006	Download QR label for the new item.	Print view shows QR image and item details.
ST-07	UC-007	VO scans QR of 'Desktop PC'.	Item card shows name, room, last-verified date.
ST-08	UC-008	VO marks 'Desktop PC' as Verified.	VERIFICATION row inserted; success toast shown.
ST-09	UC-009	Admin generates the overdue report.	Items not verified in 180+ days are listed.
ST-10	UC-010	Admin views dashboard.	Cards and charts show accurate live counts.

4. Sprint Review & Project Retrospective

4.1 Sprint 5 Completed Stories

- US-22 through US-26 — All reporting user stories completed.
- T-23 through T-30 — Dashboard, reports, CSV export, system tests, deployment.

4.2 Project Summary — Sprints 1–5

Sprint	Summary
Sprint 1	Auth, DB schema, RBAC — project foundation complete.
Sprint 2	Department and Room management CRUD — navigation shell established.
Sprint 3	Inventory CRUD and automatic QR code generation — core data layer complete.
Sprint 4	QR scanning and verification workflow — audit trail established.
Sprint 5	Reporting, dashboard, system testing, and deployment — project delivered.

4.3 What Went Well (Overall)

- Modular sprint structure kept each two-week period focused and reviewable.
- DB transaction pattern introduced in Sprint 3 was cleanly reused in Sprints 4 and 5.
- Reusing the navigation shell across all sprints avoided significant duplicated effort.
- All 10 use cases were delivered within the planned 5-sprint timeline.

4.4 Future Enhancements

- Mobile application support for QR scanning (React Native or Progressive Web App).
 - Email reminders for items approaching the 6-month verification deadline.
 - Cloud deployment (Railway or Render) for access without local setup.
 - AI-based anomaly detection for missing or misplaced inventory items.
 - Dashboard analytics with export-to-PDF support for audit submissions.
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